

Mesh refinement GUI

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Introduction:

This document is an overview of the Geomec mesh refinement functionality.

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Notes on version 3.0

Screenshots:

All the screenshots showing the main mesh refinement dialog have been updated.

Gradient values:

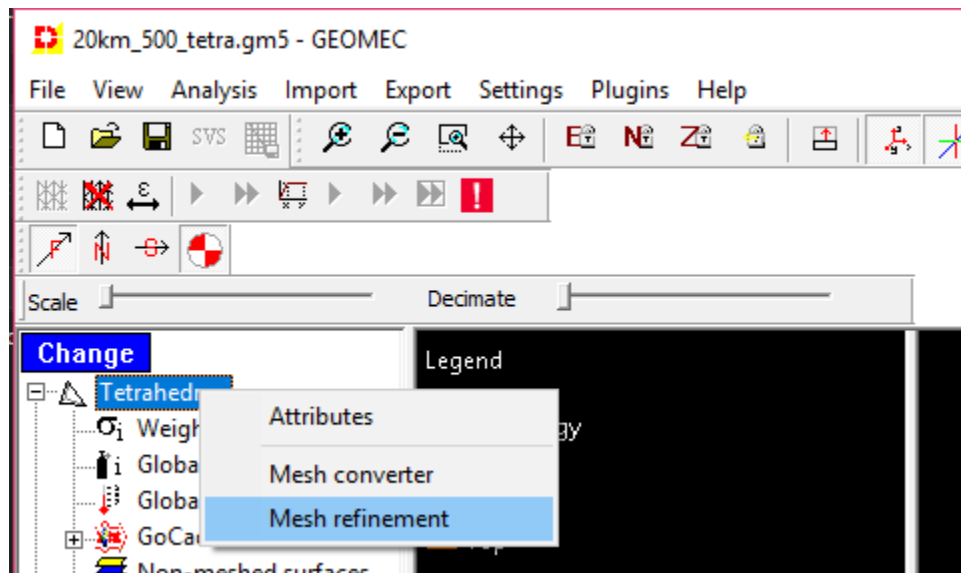
In previous version gradient allowed values were any positive integer including 0.

Now allowed values are decimal from 0.1 to 1 (can be one or two decima positions).

‘Buttons functionality and context help’ section:

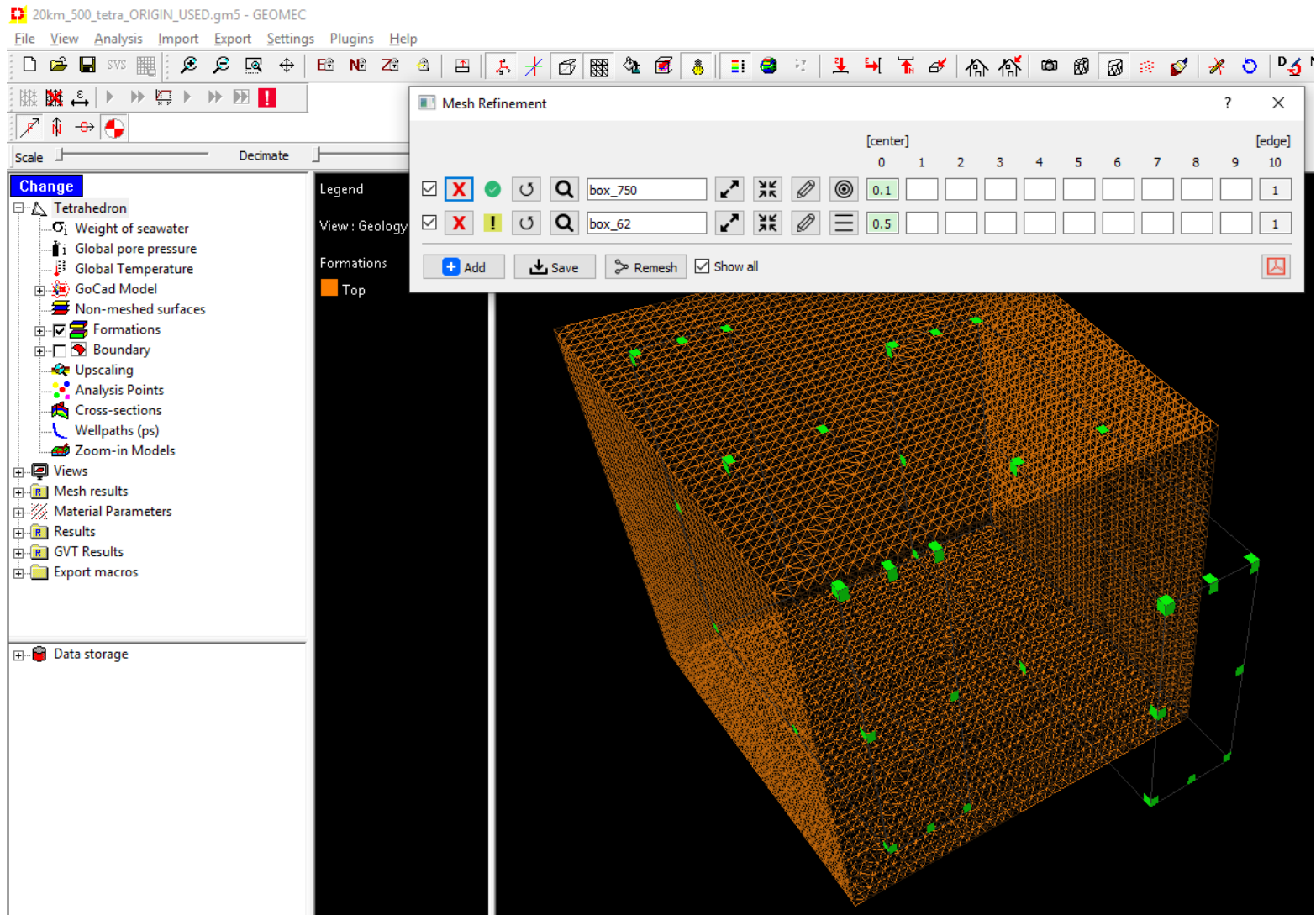
Updated to explain new functionality

Launch mesh refinement GUI:



How to access to mesh refinement GUI

GUI overview:



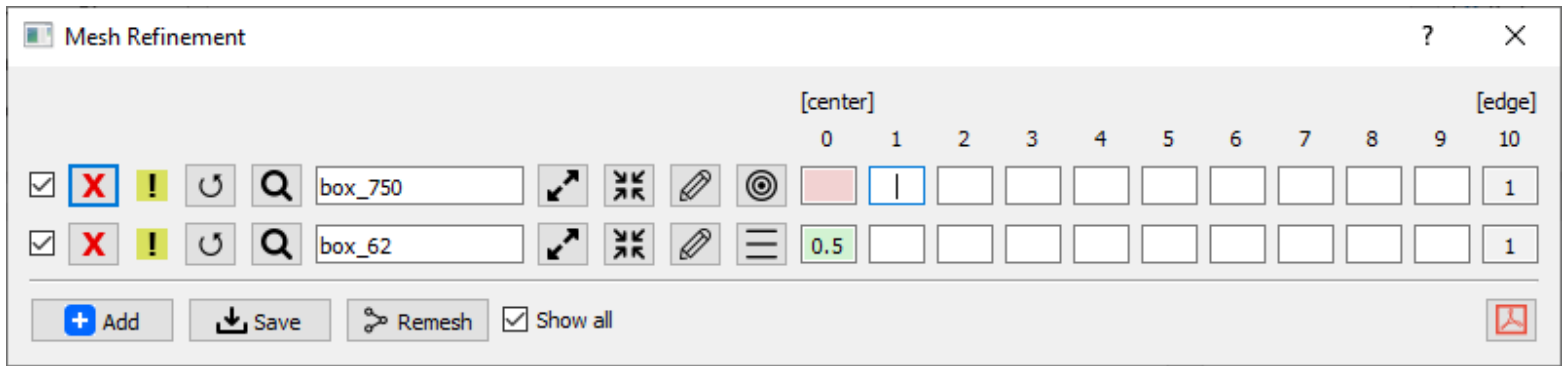
Mesh refinement dialog defining two areas to refine.

The dialog shows the boxes definition.

OIV scene shows the boxes overlapped on the model.

Size and position can be set both: from the dialog (pencil 'icon') and from the OIV scene itself

Input data validation:



The image shows a 'Mesh Refinement' dialog box with a title bar containing a question mark and a close button. The dialog is divided into two main sections: '[center]' and '[edge]'. The '[center]' section has a header row with indices 0 through 10. Below this, there are two rows of input fields. The first row is for 'box_750' and the second for 'box_62'. Each row has a series of icons: a checkbox, a red 'X' icon, a yellow warning icon, a refresh icon, a magnifying glass icon, a text input field, a zoom in icon, a zoom out icon, a pencil icon, a target icon, and a color selection icon. The color selection icon for 'box_750' is red, and for 'box_62' it is green. The '[edge]' section has a single column of input fields, all containing the value '1'. At the bottom of the dialog, there are four buttons: '+ Add', 'Save', 'Remesh', and 'Show all' (with a checked checkbox). A red 'X' icon is also present in the bottom right corner.

[center]											[edge]										
											0	1	2	3	4	5	6	7	8	9	10
☒					box_750																1
☒					box_62																1

Add Save Remesh ☒ Show all

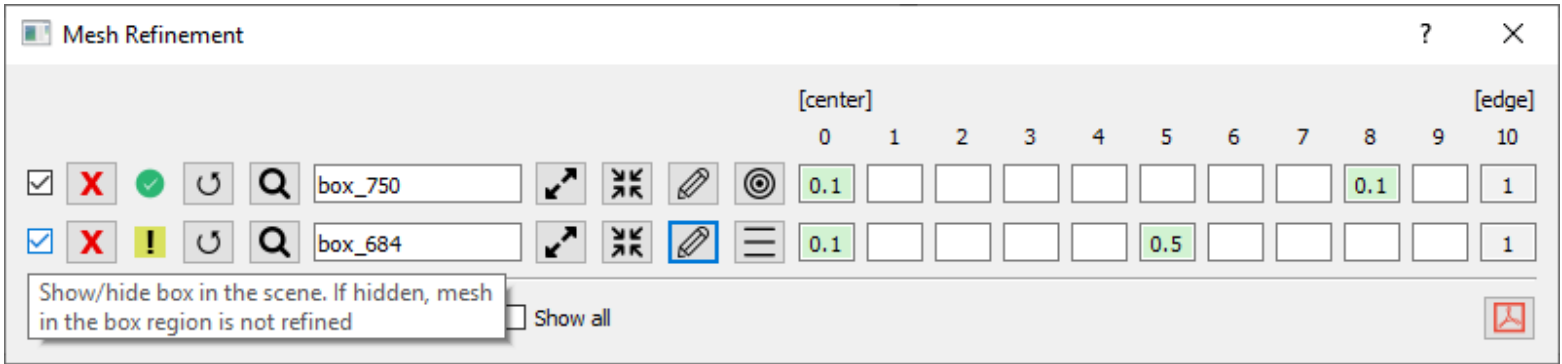
Elements relative size at box 'edge' is preset to '1'.

Elements relative size at box 'center' is **mandatory**.

Red background in first box indicates that a mandatory field is empty

Warning icon indicates something is wrong

Buttons functionality and context help:



Context help on all icons and buttons

Next a list of icons with its functionality and context help:

Name:

'box_750' Box name. It can be edited

Boxes size:

↗ Enlarge box x10 times

✖✖ Decrease box size x10 times

 Set box size and position

↶ Revert changes on box size and position

Boxes visibility:

☒ Show/hide box in the scene. If hidden, mesh in the box region is not refined

Q Zoom to the box

Add / Delete a box:

 Add a new refinement box

X Delete the box

Gradient:

Scale goes now from 0. to 1, before it could be any integer

Edge gradient is always 1 so no ediatble

Gradient values not correct, value at center or at edge not informed

Geometry validator:



Box geometry valid

Invalid: boxes overlapping and box out of scope of the model

If invalid, mesh refinement can't be performed

Changes control:



Box modifications saved

Modifications: box geometry (size and position), gradient, type (CenterPoint or PlanarZ)



Box modifications no saved yet

So not considered for the mesh refinement

Mesh refinement type:



CenterPoint type

Toggle button no switch from CenterPoint to PlanarZ



PlanarZ type

Save changes button:



Save

Saves all boxes GUI modifications so current state will be considered in mesh refinement process

Remesh button:



Remesh

It does:

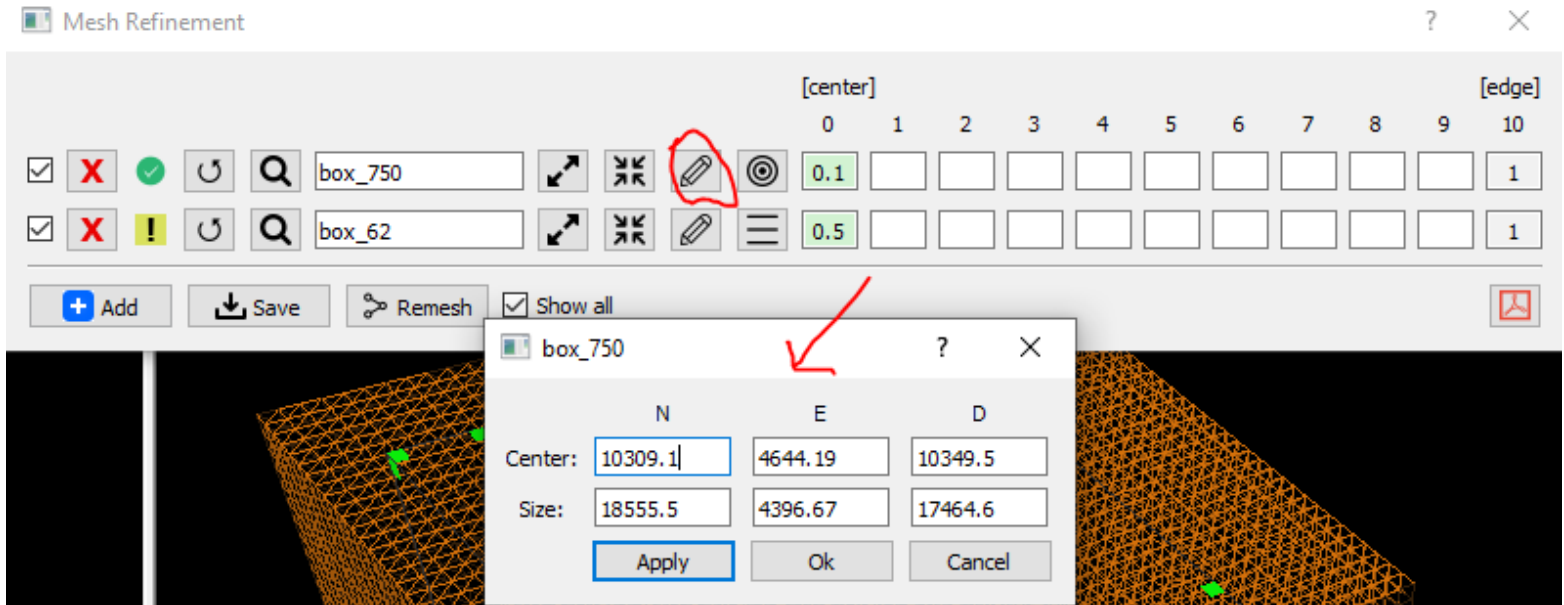
1. Saves modifications
2. Invalidates mesh
3. Create mesh

Link to manuel manual:



Link to this manual

Set size and position (form input option):



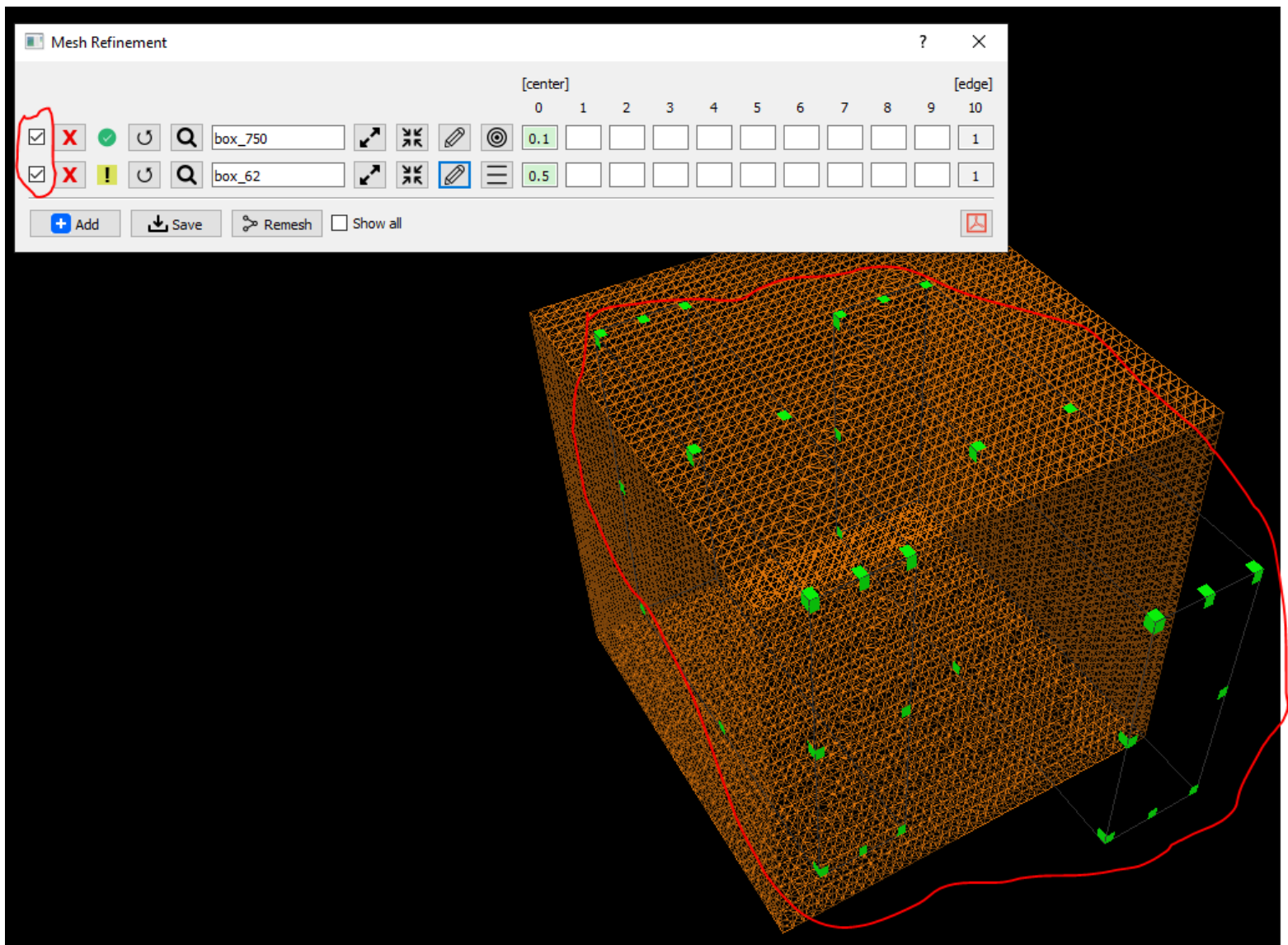
Pencil icon to set the box exact size and position

Size and position can be also be set in the OIV scene manipulating the graphic objects (boxes)

The screenshot shows the 'Mesh Refinement' dialog box with two rows of refinement boxes. The 'box_684' row is highlighted in yellow. A red circle highlights the close button (X) in the top right corner. A red arrow points to the 'Keep changes?' dialog box, which is overlaid on the main dialog. The 'Keep changes?' dialog has 'Yes' and 'No' buttons.

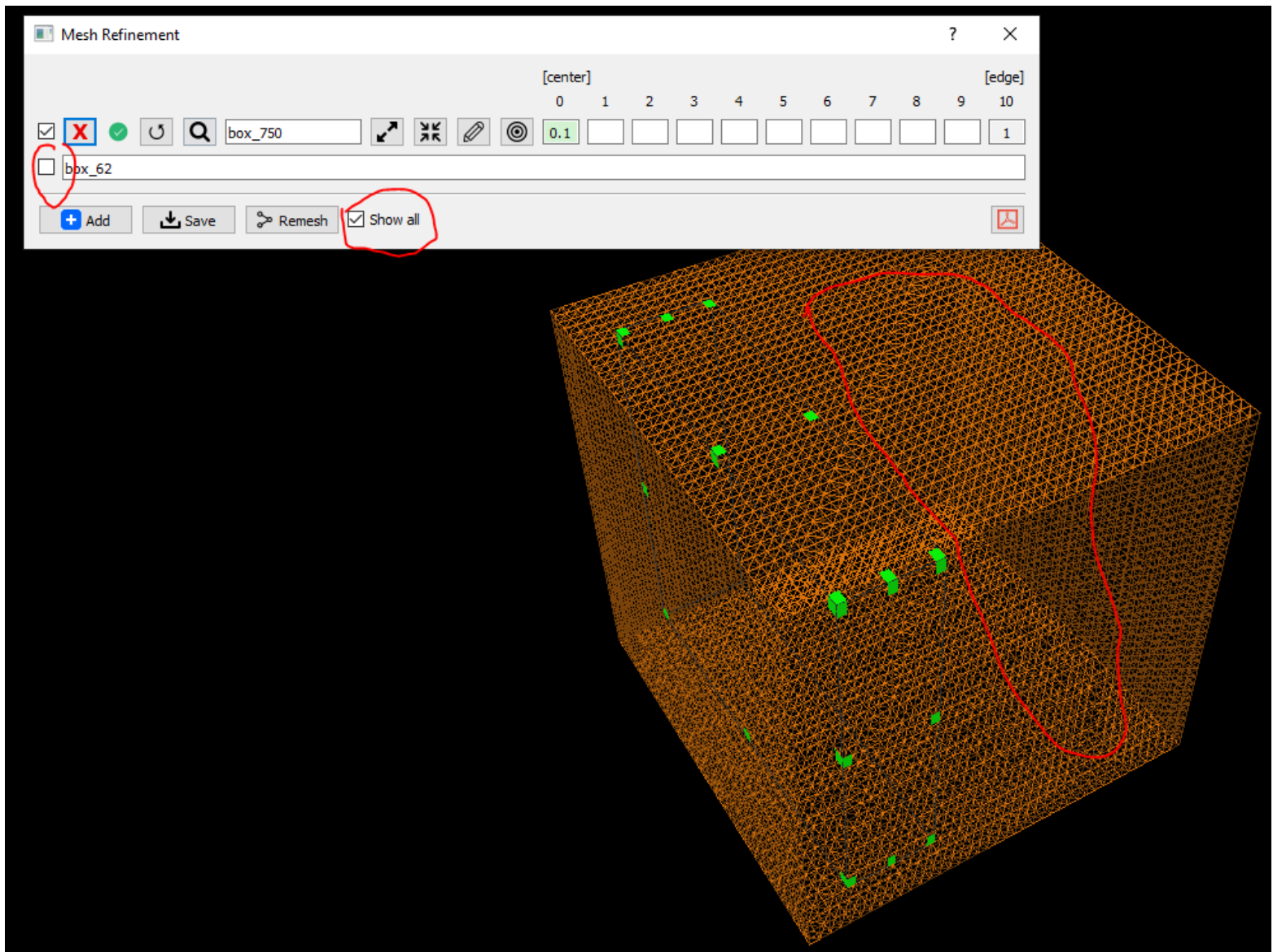
9

Enabling and disabling boxes:



Hidden refinement boxes won't be considered for the mesh refinement

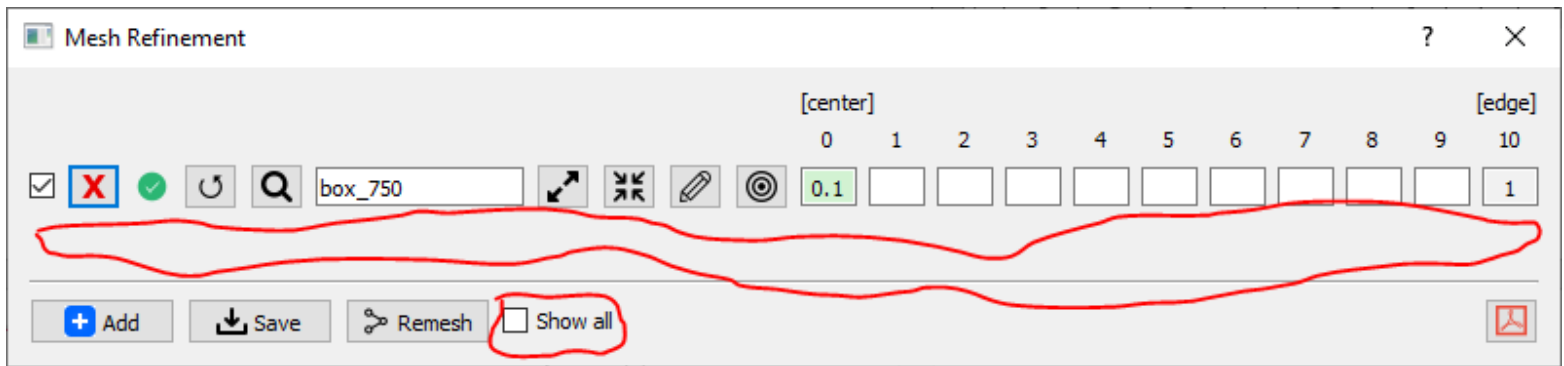
In this case both boxes are checked so both will be considered



Hidden boxes can be shown checking the 'Show all' checkbox

In this case both are shown in the list but only one (the visible one) will be considered for the refinement

Show hidden boxes allows to change its visibility



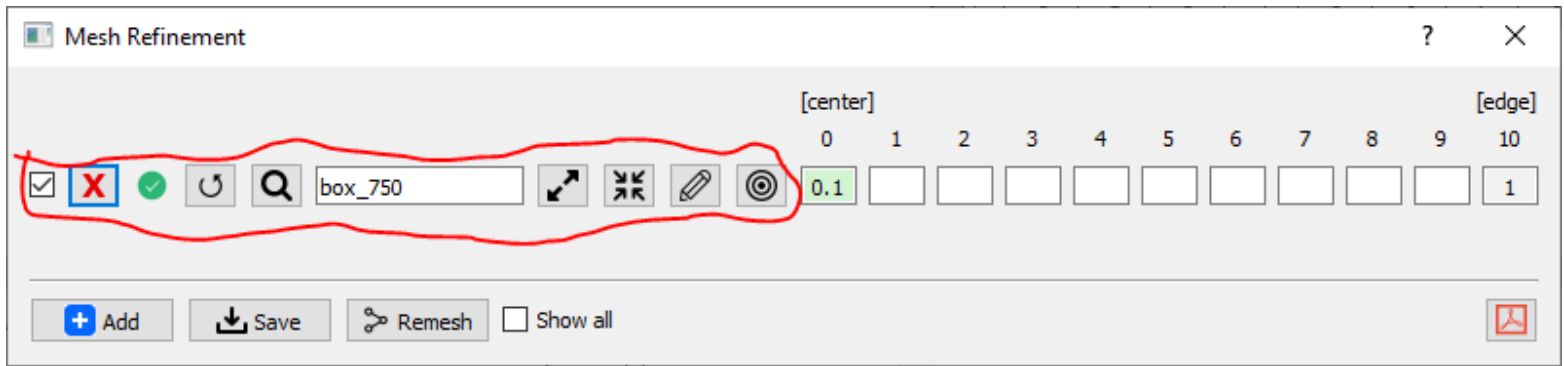
If many boxes defined

but only few of them wants to be consider to the mesh refinement

uncheck 'Show all' checkbox

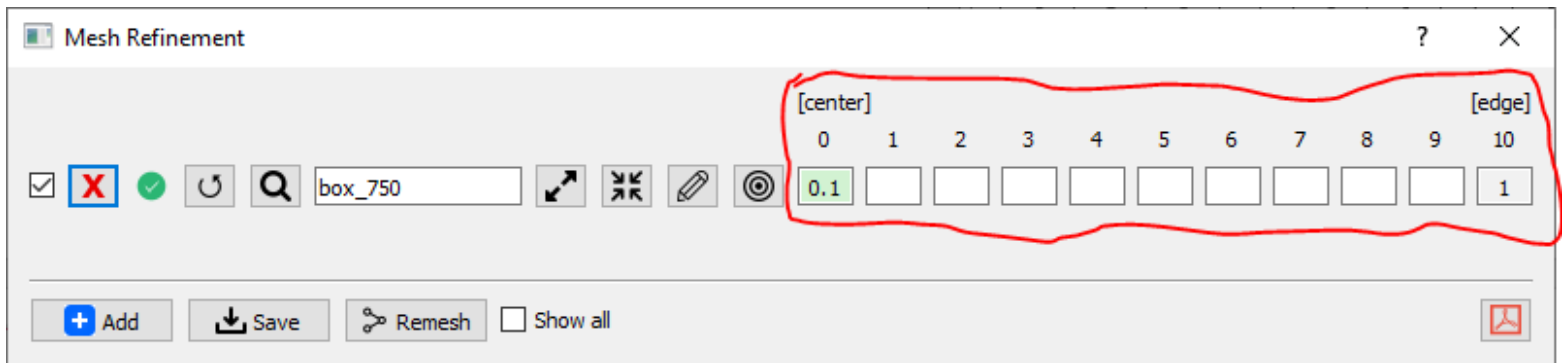
This way only boxes to be considered will be visible and the rest will remain hidden

Two parts on de box definition:



Part 1: Icons to perform actions on boxes

Icons functionality has been explained in 'Buttons functionality and context help' section



Part 2: How to define the elements size.

Size is relative.

Example explanation:

Elements at the center (**0**) (value 0.1) will be 10x smaller that those at the edge (**10**) (value 1)

Elements just in the middle (**5**) (value 0.5) will be half the size of those in the edge (**10**) (value 1).

This is because value 0.5 at the middle is half of 1 (value at the edge)

Gradient:

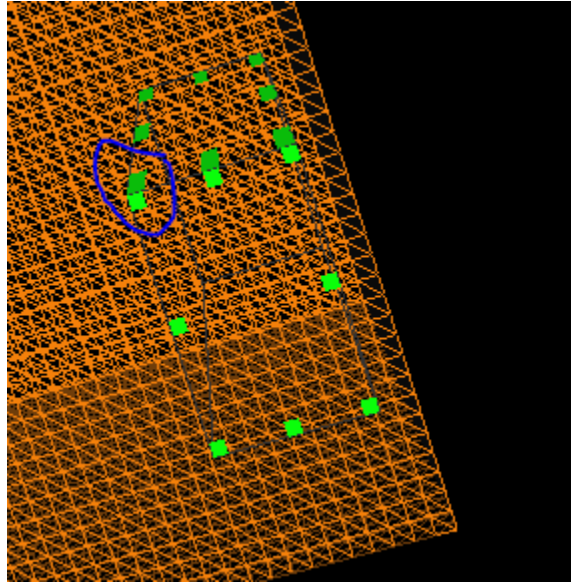
Elements in between with size values not set (1,2,3,4... 6,7,8,9) will have sizes proportional to the distance to the elements whose size has been set (1,5,10)

Comment:

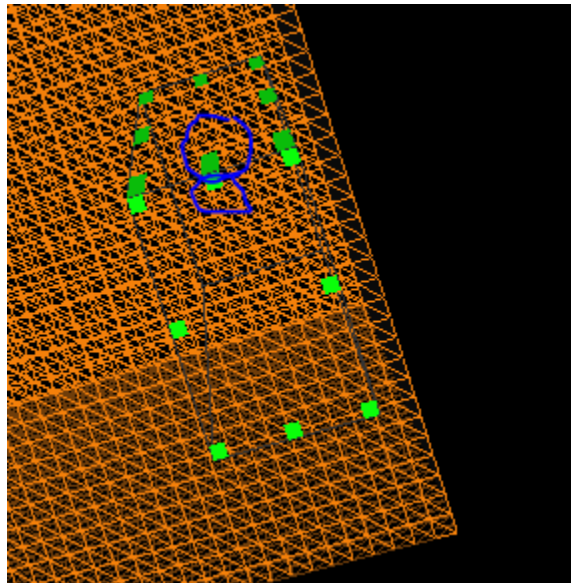
In the example above sizes '1-5-10' has the same meaning that '2-10-20' or '4-20-40' (sizes are relative)

The reference size is always the size of the elements in the edge.

Box size graphically set:

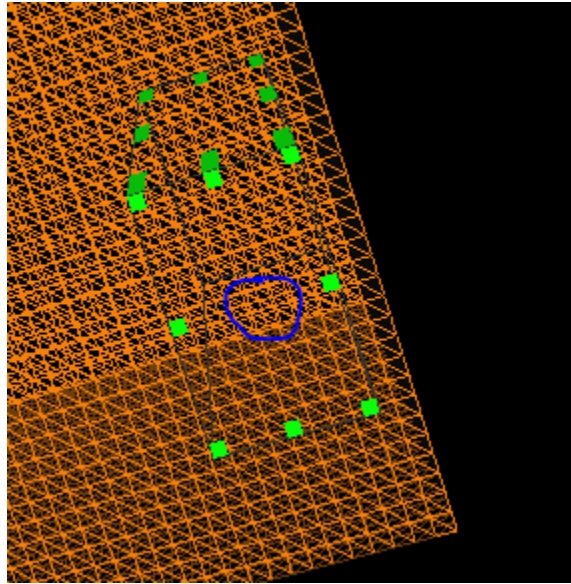


Box size can be modified in two directions at same time
with the small green boxes on the corners



Box size can be modified in only one direction
with the small green boxes in the center of the edges

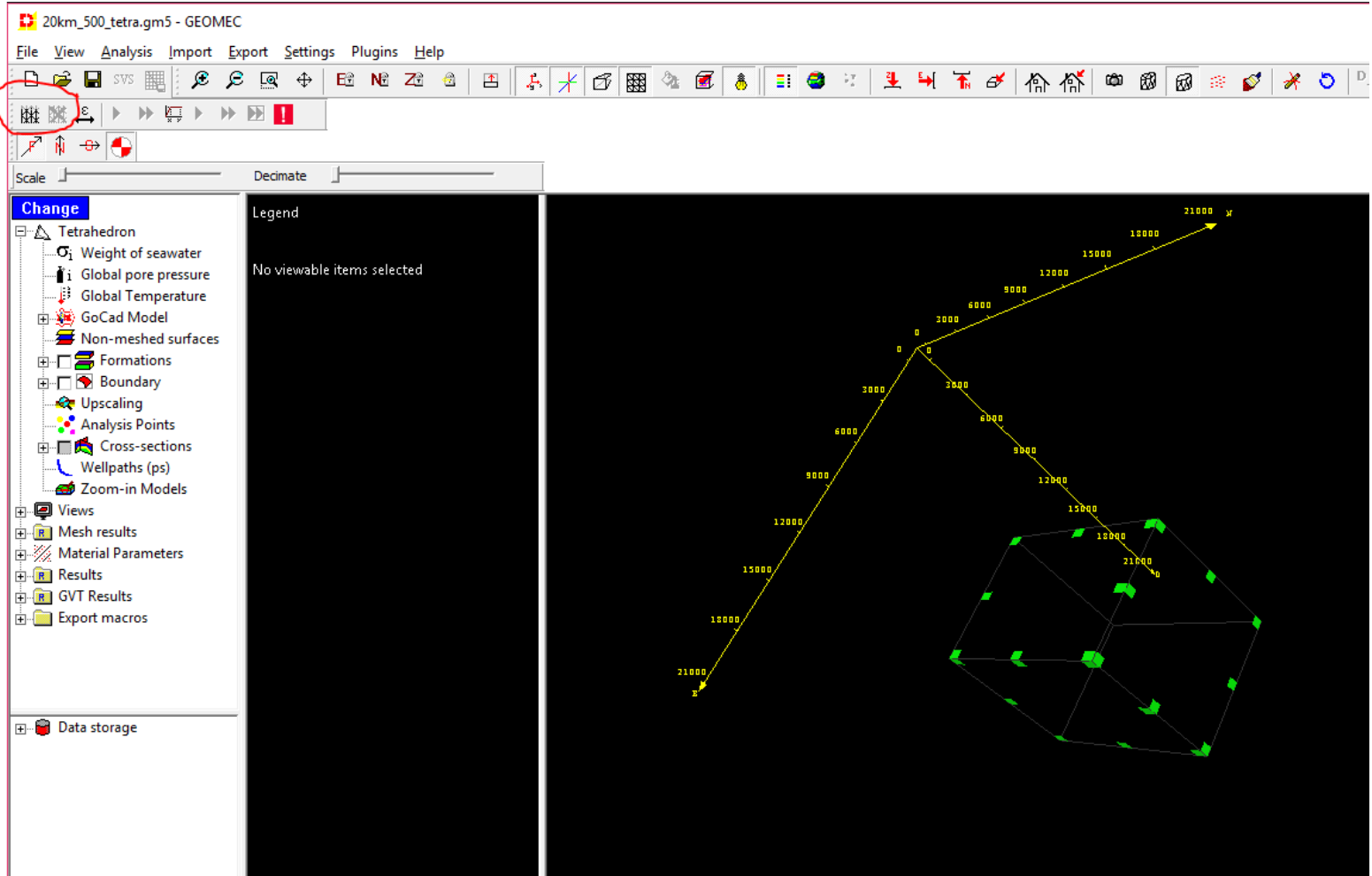
Box position graphically set:



To move the box, click on a face and hold and then move the mouse.

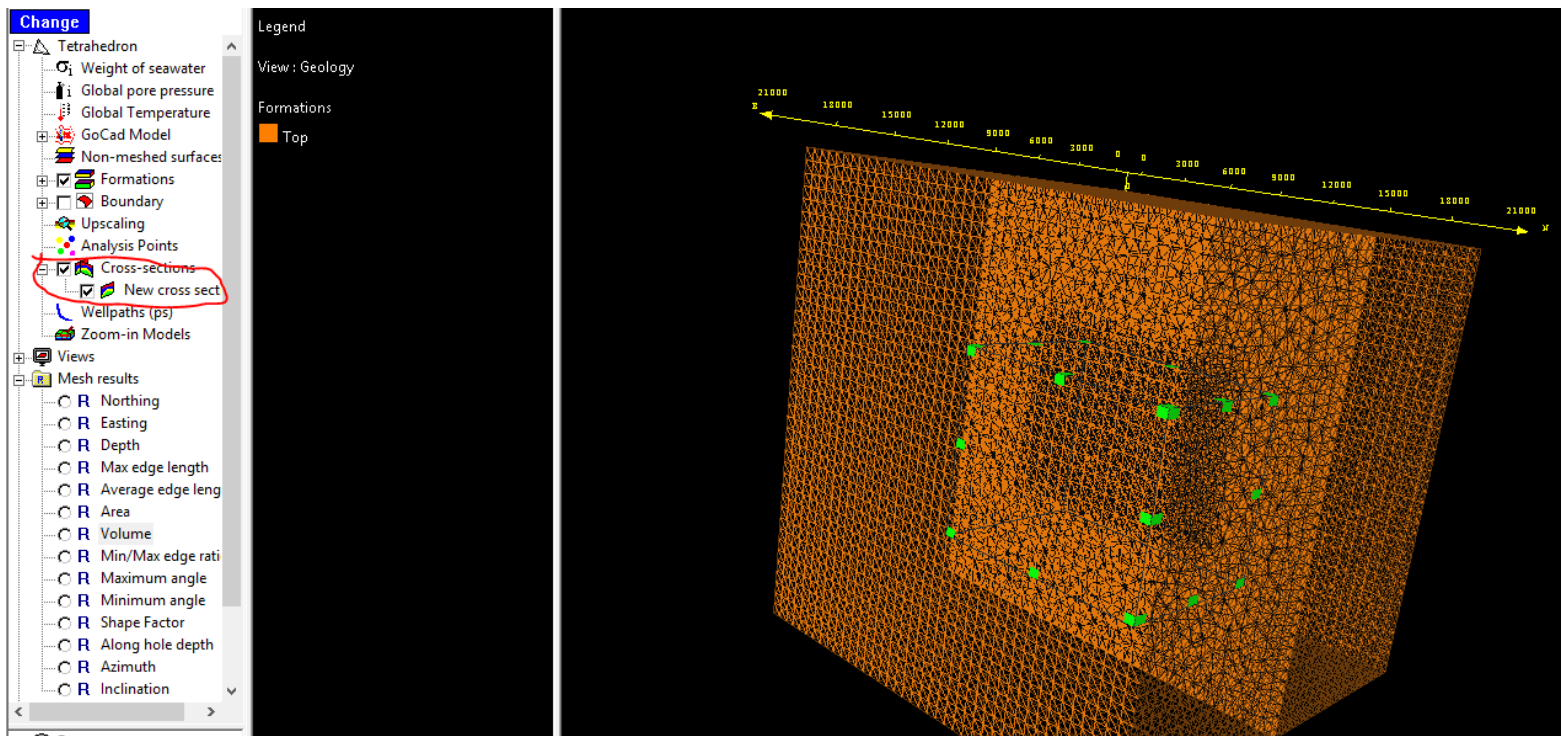
Doing this in different faces allows to move the box in all three directions

Refining mesh after having defined refinement boxes:

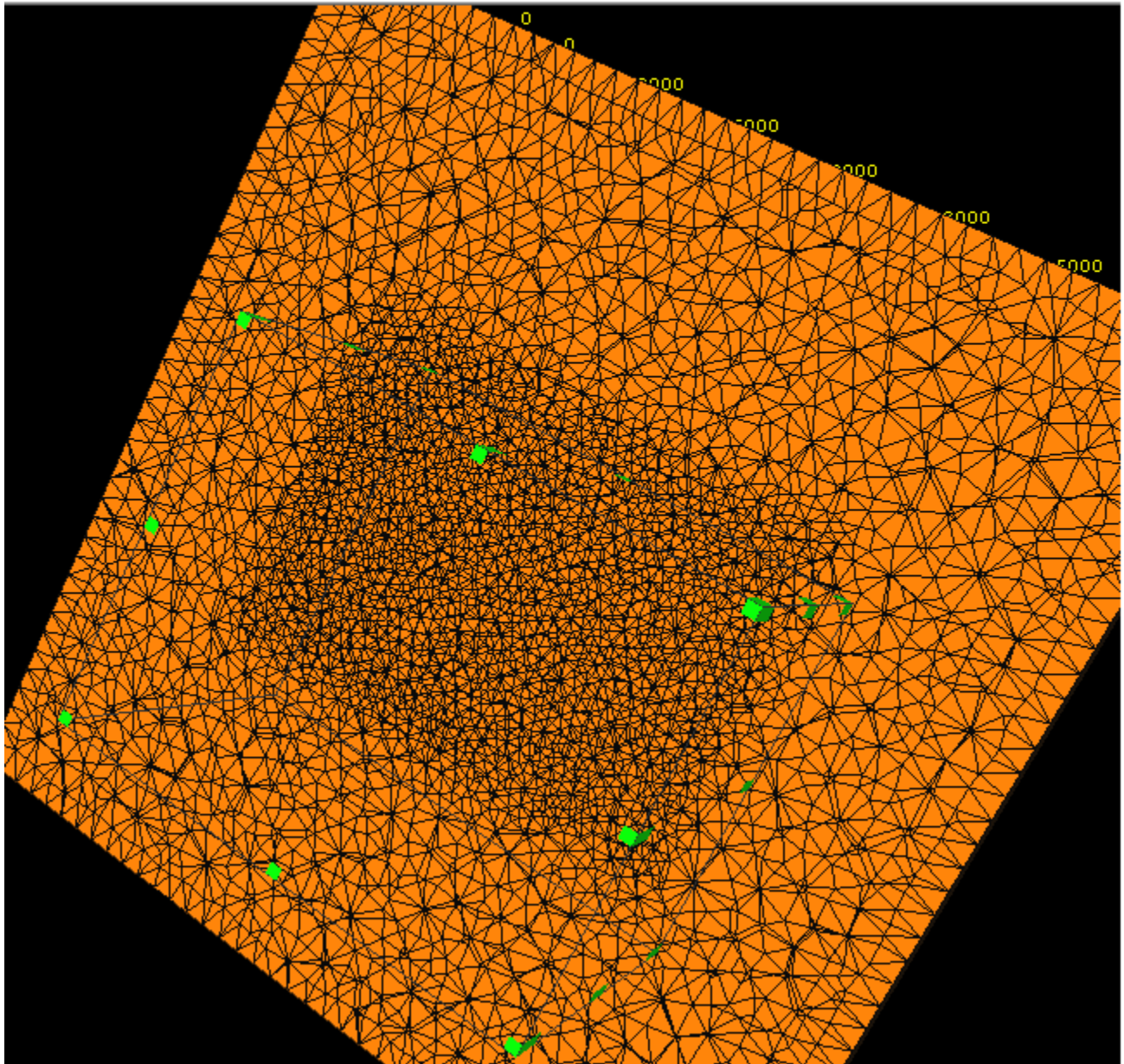


To consider the mesh refinement in the selected box a re-mesh needs to be performed

Viewing results after refining parts of the mesh: cross-section

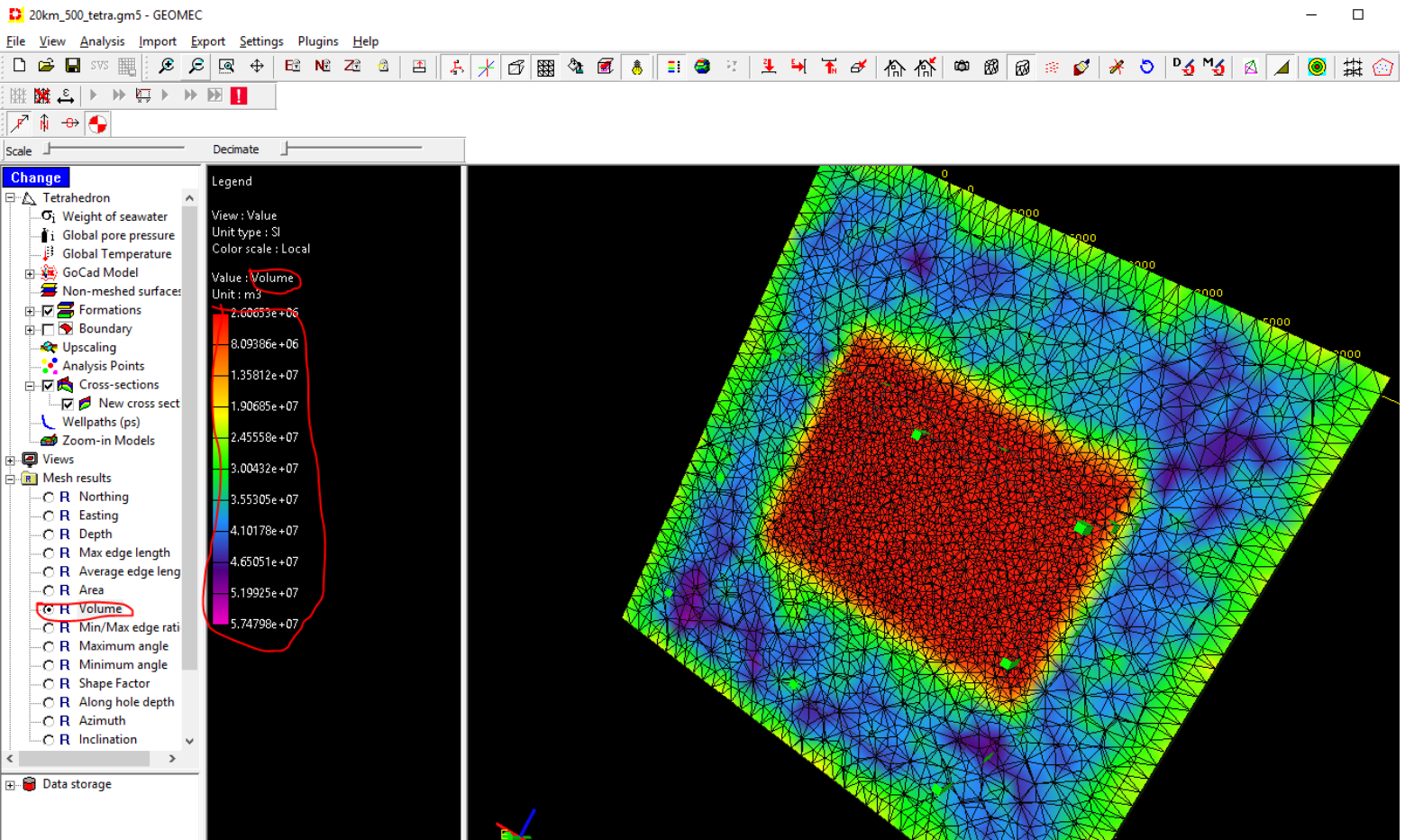


Cross-sections allow to see mesh refinement results



Clip the cross-section

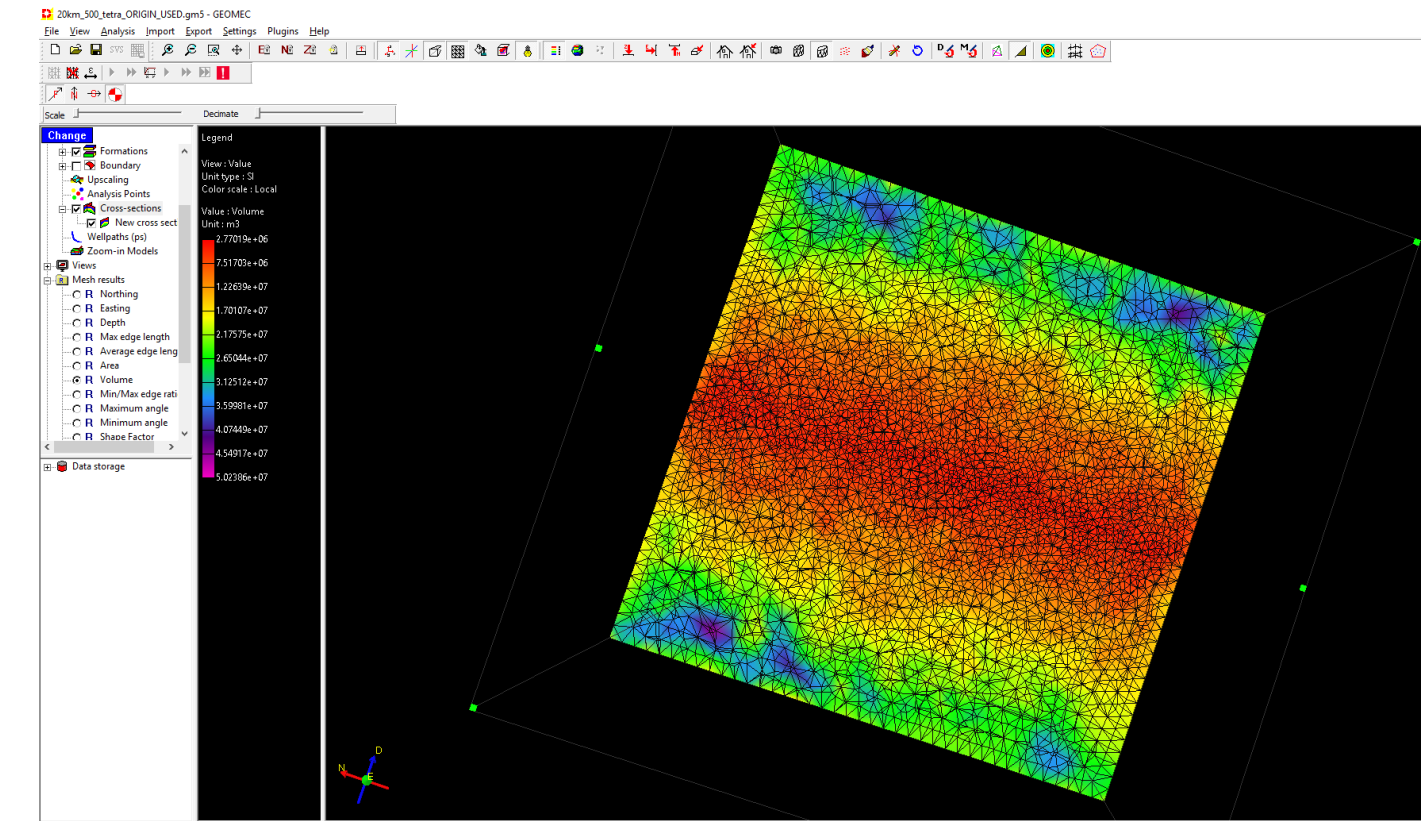
to see more clearly the difference size of the elements inside and outside of the refined volume



To view the refinement results even more clearly clip the cross-section and select 'Mesh results > Volume'

Mesh refinement type examples:

1. PlanarZ:



2. CenterPoint:

